# LIBRA Liquid Immersion Blanket - Robust Accountancy

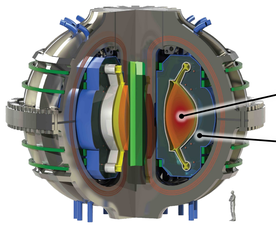
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## Tritium breeding

2030



The ARC concept relies on the Liquid Immersion Blanket (LIB) for breeding tritium

$$D + T \rightarrow He + n$$

$$Li + n \rightarrow T + He$$

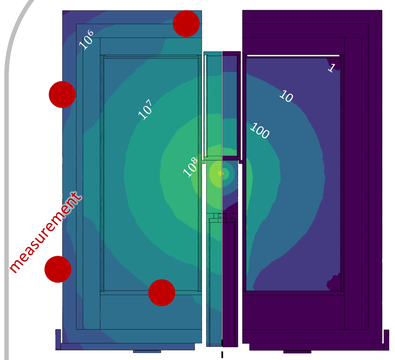
$$\text{Tritium Breeding Ratio} = \frac{\text{Tritium produced}}{\text{Tritium consummed}}$$

Can we test this concept now?

### Goals of LIBRA

- To demonstrate the concept of LIB (TBR ≈ 1)
- To improve the Technology Readiness Level of LIB
- To validate numerical models for tritium transport
- To gain knowledge on molten salt behaviour in fusion environments

## Neutronics




Neutron flux $n/cm^2/s$ | Tritium production $nCi/l/h$

- Neutronics simulations track the path of neutron particles
- Used to predict tritium generation, neutron flux, heating, production of activated materials…

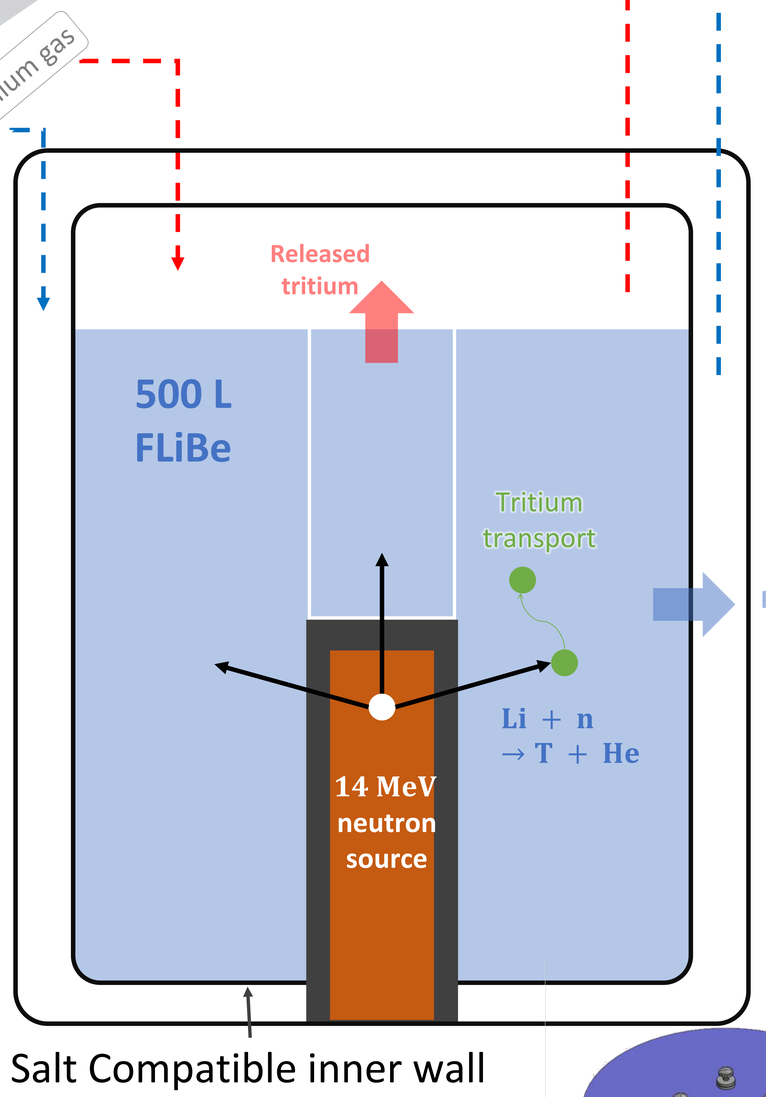


Activation foils

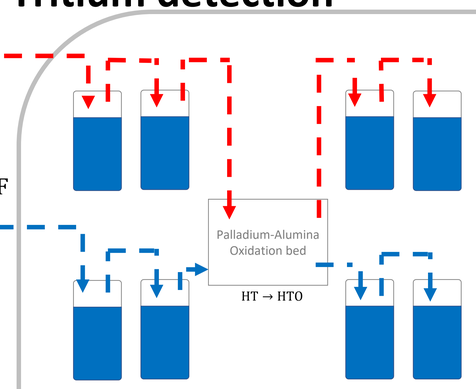
- LIBRA will provide neutronics data for validating codes
- Activation foils, liquid organic scintillators and diamond detectors as used to characterize the neutron source and the experiment



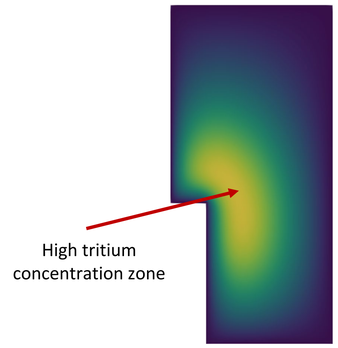

Diamond detection




## Tritium detection



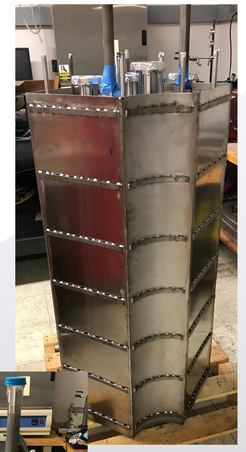

- Bubblers with oxidation furnace capture different chemical forms of tritium
- Salt radiochemistry impacts tritium forms and transport dynamics
- LIBRA will provide data for simulation codes validation
- The impact of sparging bubbles in the FLiBe will also be studied

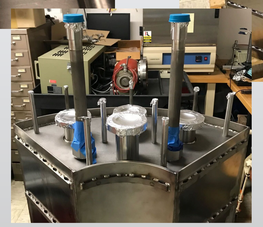


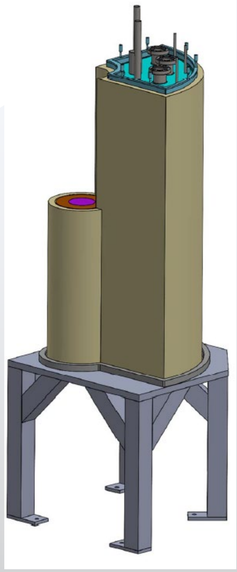

**Tritium transport can be modelled at full scale**

## LIBRA Pi Sector



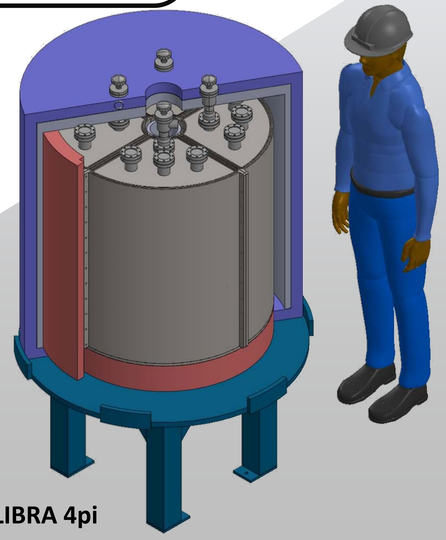






## LIBRA 4pi specs

| | |
|---|---|
| Salt volume | 500 L |
| TBR | ≈ 1 |
| Temperature | 700 °C |
| DT Neutron energy | 14 MeV |



LIBRA 4pi

### Higher fluid temperature = Higher Efficiency

